



Retention of Critical Knowledge - Beth Collier – Environmental Specialist: Air, Water and Waste - GSUSI-PTS-S

Retention of Critical Knowledge Report

Shell USA, Inc.

SRN-010382

September 2024

Author(s): Kosiara, Mark SHLUSA-HRL/LK; Collier, Beth B GSUSI-PTS/S

Executive Summary

The ROCK exercise for Beth Collier was undertaken because Ms. Collier was retiring from Shell after 21 years of Shell service. This ROCK was intended to supplement the handover work that had already been done between Ms. Collier, her team lead, Rob Smith and the team member taking her position, Maurice Pinder. Given Ms. Collier's extensive experience in managing the Environmental categories of air, water, and waste at the Shell Technology Center Houston (STCH), multiple items that have been captured in this report will be valuable to those working in this space in the future.

The initial ROCK interview was held on May 21st, 2024. Unlike other ROCK reports, this ROCK intended to capture an inventory of the various regulatory jurisdiction and reporting requirements for the key Environmental media (Air, water, and waste) that were managed by Ms. Collier as well as any historic Environmental information that would not necessarily be common knowledge at the site due to Ms. Collier's tenure in this role. In addition to this, the report also contains the more classic goal of capturing: key lessons learned, best practices, risks/challenges for the future, and recommendations/opportunities for improvement in the various topics discussed during the interview. This report resulted in the latter items being captured during the ROCK interview:

Category of Finding	Number
Best Practice (Desire to Replicate)	3
Lessons Learned (No Specific Desire to Replicate)	2
Risks / Challenges for the Future	4
Recommendations for the Future / Opportunities for Improvement	2

These above items are detailed in the main text of this report in line with the interview agenda, but they are also summarized by category in Section 9-12 of this report. For your information, you can see the detailed agenda for the ROCK interview in the agenda flow diagram embedded below.



Final Agenda.jpg

Contents

Executive Summary	2
1. Brief History of Time in Role and Key Roles & Responsibilities	5
2. Regulatory Framework (Air, Water, Waste)	6
2.1. What are the applicable laws and requirements here?	6
2.1.1. Air	6
2.1.2. Water	7
2.1.3. Waste	11
2.2. How does the information flow to meet the regulatory needs?	12
2.3. Processes at STCH for compliance and permitting new sources?	13
2.4. Key Stakeholders at STCH (20+ VP-ships & Multiple Project Groups)	14
2.4.1. Project Groups – Capital Projects, Real Estate, 3 rd Party Tenants	14
2.5. Regulatory Contacts & External Contacts (Key Contacts)	15
3. License to Operate Success	17
4. Site Specific Topics (Brief History & Current State of Each)	19
4.1. MB Tank Farm, Contamination	19
4.2. Holding Ponds	19
4.3. Well #3 Issue, Well #1	20
5. LOPC Reduction Campaign	22
6. Sustainability: Net Zero Emissions, Respect for Nature, etc.	23
7. Environmental Support Work (Consultation, MOCs, Project Recycling, Respect for Nature, Flaring Issues, Chilled Water Pump)	25
8. Personal Networks	27
9. Best Practices	30
10. Lessons Learned	32
11. Risks / Challenges for the Future	34
12. Opportunities for Improvement / Recommendations for the Future	37

13. Key Information Sources	38
Literature References	39
Bibliographic Information	40

1. Brief History of Time in Role and Key Roles & Responsibilities

Ms. Collier has been at Shell for the past 21 years. At Shell Technology Center Houston (STCH) she has been responsible for managing the air, water, and waste scope within the environmental segment of HSSE.

Key Roles & Responsibilities

Within each of the three areas there are responsibilities for:

- Record keeping.
- Reporting – for example into the Texas Commission on Environmental Quality (TCEQ) or to Shell Group.
- Compliance with permit and associated regulatory requirements that are linked to permit requirements.
- Other general responsibilities including site operations and site project support to ensure alignment and compliance with all applicable Environmental and permit related requirements.

2. Regulatory Framework (Air, Water, Waste)

2.1. What are the applicable laws and requirements here?

2.1.1. Air

Air management at the site has many associated permits, which include:

- A Title 5 permit as well as several permits by rule (PBR). PBRs are general operational permits that are relatively easy to obtain if operations meet criteria.
- The different PBR permits at STCH are associated with:
 - 2 boilers, which are a Standard permit for Boiler 1 and a New Source Review (NSR) permit for Boiler 2.
- The cogeneration unit which has a PBR permit, which is a general permit for turbines.
 - The microgrid which has a standard permit associated with the (stationary) natural gas generator, which is a non-emergency generator.
- The other non-emergency generator is associated with the laboratory at MS and can be run at any time (under this PBR). Special requirements exist if exceeding a specific number of run hours.
- Most buildings have an emergency generator, which are under restriction regarding their use (all are PBR permits).
- Painting
- Abrasive blasting
- Welding
- Degreasing
- Flare
- See list of PBRs in Environmental files.

Air Reporting Requirements

Title 5

Title 5 is a federal operating air permit administered by the Texas Commission on Environmental Quality (TCEQ). It has a semi-annual reporting requirement associated with it, which is called a Deviation Report. These reports need to be signed off by the Site Signatory Authority, who for TCEQ purposes is considered the Regulatory Official (RO) for these two reports.

Additionally, there is an annual Certification Report that also needs to be signed off.

Air Emissions Inventory (AEI)

There is an annual Air Emissions Inventory report, which includes an overview of all the

emissions from all laboratories, the boilers, CEB/flare, cogeneration unit etc. The site air fees are based on those emissions each year.

This AEI report is the basis for several other annual reports:

- Greenhouse Gas Emissions report – a federal report submitted through the e-GGRT system.
- Annual NO_x certification/Mass Emissions Cap & Trade (MECT) – STCH is allotted a certain volume of NO_x emissions, any excess can be traded or sold. Management has decided to keep these emissions credits in the event that STCH needs to expand operations in the future, they could use these emission credit for these new emission sources.

Texas Tier 2

The Texas Tier 2 report is an inventory of maximum annual storage. It is the maximum inventory of a chemical on any specific day kept on site throughout the period of 1 year. This falls under the Community Right-to-Know Act and this information gets submitted into the State of Texas Environmental Electronic Reporting System (STEERS).

There are minimum thresholds associated with the hazardous chemicals included in this inventory and associated report. If the threshold for a particular chemical is not met, then a chemical does not need to be reported.

Aside from the TCEQ, the Tier 2 report is shared with:

- The local fire department, and
- The Local Emergency Planning Committee (LEPC).

This allows those organisations to plan for a response to a possible emergency on site.

Fire Extinguisher Training

The Fire Extinguisher Training report is written at the start of each year. This report lets the city know if the site is planning to do any open burning and it is linked into the schedule for fire extinguisher training that the fire chief and site training coordinator maintains.

Fees

Finally, there is a responsibility to report any fees incurred associated with the issuing of any of the previously mentioned reports. A tracking document in Assurance SharePoint is used to monitor annual fees and associated payments.

There are various types of fees paid for Environmental permits, applications for permits, for emissions released, for waste generated, for violations if they occur and decide to cite us.

2.1.2. Water

There are several different types of water at STCH:

- Potable water.
- Groundwater.
- Storm water.
- Wastewater.
- Steam and process water. Note: Utilities staff deal with this category of water and Ms. Collier has little responsibility for management of this category except for when it falls under the loss of primary containment reporting requirements or becomes a sustainability issue. There are Shell Environmental Reporting requirements for reporting water consumed at the site.

If there are any reportable incidents, then the first 4 categories above need to be reported under the incident reporting system under the Environmental category.

Water Reporting Requirements

There are several agencies to which water use needs to be reported:

- Texas Water Development Board – An annual report is required that includes a survey of ground and surface water use, which includes potable water received from the city of Houston, as well as water pumped up from the groundwater well on site.
- Harris-Galveston Subsidence District (HGSD) – HGSD provide the permit for the groundwater well. It is well #2134, better known as Water Well 3 on the site.
 - An annual report on water pumped is required by HGSD. The City of Houston supplied water goes on this report as well.
 - An annual permit needs to be obtained for Well 3. An application for permit is submitted which starts the renewal process.

The permit has requirements around calibration of the flow meter. STCH has two meters that are swapped out annually and calibrated annually. Calibration is done by third parties and the third party submits the calibration report to the TCEQ. While the requirement to calibrate is not as frequent as every year, this practice is followed to avoid missing this activity as has happened historically.

STCH submits a Meter Card Report annually to let HGSD know what meter is in service, as required.

- The Safe Drinking Water Act, managed by the city of Houston drives backflow preventor compliance at STCH. The backflow preventer protects the city from any possible contaminants that might come from STCH's water system. It also protects each system where those preventors are installed.
 - There are domestic water preventors, fire water preventors, and irrigation preventors.

- The Brown & Root Master Plumber currently does this testing for the irrigation and domestic preventers. In the past, the American Fire Protection Group (AFPG) has also done this testing.
- The testing of the fire backflow preventors requires special certification, thus, AFPG still completes these tests.
- Once the test is completed, it is entered into the SWIFT compliance system or current system the City of Houston is using (SWIFT is Houston's programme to manage due dates and fees). [Ms. Collier does not have direct access to the SWIFT system, so she is dependent upon a report produced from the SWIFT system, which usually lags compared to when reports have been submitted. Subsequent checking on SWIFT is therefore required to verify compliance with reporting requirements.]

Potable Water Testing

- Potable water Residual Chlorine testing and flushing is done by the SET group. The city should be sending the site chlorinated water, but water is treated differently at various city treatment locations and on occasion the site receives water with low chlorine residual. SET tests for both types of chlorine residual at the sample point where water comes into the facility. If there are issues, it can be managed. The chlorine residuals in the storage tank on site are monitored for correction. If issues are found, then the city of Houston is notified. This could also require that the internal water lines at the site need to be flushed out. (if the issue was detected too late) There is a policy in place that details the potable water procedure. It is in the HSSE Manual and covers everything from the regulatory levels that chlorine residual must meet, and it lists the flushing and testing points. (See HSSE Manual Potable Water Procedures.)
- Although Ms. Collier is not directly responsible for the execution of this work, she does track compliance because she must manage any reporting and inspection activities associated with compliance to the regulations.
- Note: Permits are posted in the STCH Assurance SharePoint site. Each media has a permit file.

Storm water requirements

There are several key components to storm water:

- Texas Pollutant Discharge Elimination System permit. A storm water general permit, which is somewhat like a PBR. If the criteria of the permit are met than anyone can have that permit, which is better than having individual permits for each sector of the facility. It is a multi-sector general permit for industrial facilities.
 - Storm Water Pollution Prevention Plan (SWPPP) - One of the requirements of the permit. This plan lists all the potential areas of contamination. Sources that

are permitted to go into the storm water system are identified. Testing requirements are also listed. If all these requirements are kept in place, then the general permit can be used. If there are issues, then it might mean that individual permits are required. (leading to higher cost, higher requirements)

Annual training of individuals concerning activities (SWPPP) that could contribute to contamination of the storm water system is required. Ms. Collier does this training, and it normally involves individuals who are active in outdoor processing areas. Work is also done with the landscape crew, as well as any third-party contractors working on site. The training plan is reviewed annually. (Note: Training documents are stored in SharePoint under the SPCC and SWPPP section within the water folder.)

When STCH still had individual stormwater permits in place, after each rain occurrence, someone had to go out and take samples. Each sample had to be analysed on 4 parameters by an external lab (which had collection, shipping, analytical, reporting etc. costs). Those reports had to be consolidated into monthly Discharge Monitoring Reports (DMR). There was also a requirement to do biotesting every 5 years. The multi-sector general permit only has a yearly requirement to test for metals. This data goes into a Discharge Monitoring Report (DMR), and it is only submitted if there are failures (to date none have occurred). This general permit is on a 5-year cycle.

Best Practice – Environmental Training is more effective when done in person for on-site staff that could impact storm water discharge limit compliance

When training staff and contractors at a laboratory facility like Shell Technology Center Houston, it is important to help them understand that they are an important part of the process in keeping the storm water conveyance system as clean as possible. Each activity on site can cause issues for the storm water system if precautions are not taken. It is therefore important to follow the Storm Water Pollution Prevention Plan and make sure there is good communication (best achieved through in-person visits) with the various parties working on site.

Compliance Calendar

There is a compliance calendar that details which reports are due when and the regulations that drive each report submittal. It is a managed Excel spreadsheet, stored within Assurance SharePoint.

Best Practice – Environmental compliance requirement monitoring is improved through use of a central tracking tool:

All laboratory site Environmental compliance requirements (i.e.: testing, reporting, training) at Shell Technology Center Houston are managed together in one single electronic repository. (i.e.: An Excel spreadsheet) Centralizing this information for

the requirements of all environmental media (Air, Water, Waste) helps ensure that compliance oversight is consolidated and can be monitored/managed in one place. Distributing these requirements to various departments or individuals leads to higher exposure for non-compliance during staff/organizational changes.

2.1.3. Waste

Managing the Waste media deals with collection, handling, storage, and disposal of various waste stream categories at site. It also requires managing the different reports required for these waste streams and the associated waste management processes.

- Annual Waste Summary – A key report that SET creates based on all the on-site Hazardous and Non-hazardous class 1 materials (materials that have been a part of an industrial process but are not strictly hazardous/dangerous). After the document receives the necessary approvals and is signed off, it is submitted to the TCEQ via the STEERS system. Once it is in the system, minor adjustments can be made on an annual basis. (TCEQ State of Texas Environmental Electronic Reporting System STEERS) ()
- Waste Source Reduction Plan. Because STCH is classified as a large quantity generator of waste, there is a requirement to create a plan to reduce waste s volumes. SET also creates this plan and works with waste generators on site to identify waste streams that would be candidates for reduction. Ms. Collier oversees this process. A 5-year plan is presented to the state and progress is reported annually in the Annual Performance Report (APR).
- All wastes streams at site are tracked, even though some do not require reporting to regulators. Internal Shell requirements require a report to be generated quarterly for Shell head office.

Project Environmental Support

There are two types of projects undertaken at STCH, ones that build or remove buildings, and research projects. These projects can have various durations, so it is critical to support these projects in managing Environmental permitting requirements, etc., which is a key role of Ms. Collier's position. This also means providing training to those individuals that are required to sign off on the various waste manifests, reports, etc. associated with these projects.

Waste Facility Assurance & Verification

Another activity managed by Ms. Collier is the assurance and verification of the various facilities (e.g.: Reuse, Recycling, Disposal) that Shell uses to process waste materials. These facilities need to meet Shell's requirements. STCH attempts to limit the number of facilities it uses for waste management so that the long-term Environmental liability is reduced.

STCH has joined an industry consortium CHWMEG (Not an Acronym) that audits Offsite Residual Management facilities. This means that each member company does not have to go out and audit each waste treatment facility they use. CHWMEG Audits can be very extensive with reports being over 100s of pages, and these audits are done on a 3 to 5-year cadence. STCH Waste Leads analyse these reports and recommend facilities that are suitable for use. (Note: There are also approved waste subject matter experts (SME) within Shell that can be called upon for support when needed.) If SET is considering sending waste to a facility not currently being used, its approval status for use can be verified by consulting the list of approved facilities or an audit must be considered for the new site before use depending on the type of facility. (Note: A complete list of approved facilities can be found in the Waste Catalogue, which is managed globally by Cindy Chen.)

Class 3 Non-Hazardous Waste

It is generally not beneficial to label waste as class 3 because this classification of waste requires special additional analysis. The price difference between class 2 and class 3 disposal is not big enough to justify the additional costs associated with analysis of the waste. It would only be beneficial for very large volumes of class 3 waste.

There is a waste management procedure, found in the HSSE manual, which speaks to how each category of waste (e.g., batteries, light bulbs etc.) should be managed. The manual helps direct an employee to the right person who will know how to dispose of a specific waste.

If SET notices any miss-management of waste disposal (e.g., throwing aerosol cans in regular waste bins) then they will contact Ms. Collier to help inform the parties that are managing waste incorrectly. Follow up by Ms. Collier is intended to educate/train the individuals, so the wastes are managed properly going forward. An event should be entered if this occurs.

2.2. How does the information flow to meet the regulatory needs?

Meetings per Calendar Year

Outside of non-scheduled regular communication, there are several cadence meetings held during a calendar year to manage Air, Water, and Waste related issues:

- A notice of registration meeting is held two times a year. This meeting discusses all the waste streams and waste units on site.
- Three times a year there is a meeting on storm water to discuss any concerns and to complete the Comprehensive Site Stormwater Inspection required by the Storm Water Pollution Prevention Plan (SWPPP).

A hard copy yearly calendar is used to track compliance-related deliverables. (This was kept previously by Ms. Collier.)

SET has given their employees and others a SWAG card that can be used to stop work. This is a guarantee that any employee has the authority to STOP work. Any employee at STCH is empowered to initiate a SWAG if they feel that activities are not meeting HSE requirements. If there are any issues related to the Stop Work Authority Guarantee (SWAG), this would be dealt with in a special ad hoc meeting.

Spill Prevention Control & Countermeasures (SPCC) Plan

The SPCC plan is also a requirement at STCH. The Environmental Specialist, Ms. Collier, is responsible for this. The SPCC plan is mandated by the Oil Pollution Act (OPA) 1990 because there is a state water boundary at the northern edge of the STCH site.

The plan needs to be updated every year as needed and every 5 years needs to be certified by a PE. The plan will need to be renewed in 2026, which is normally done by a consultant (AECOM).

Documents

All Environmental documents necessary to meet regulatory and internal Shell requirements are stored on SharePoint. A large amount of data is collected from various sources at STCH, and it is then fed into the various reports. If an Environmental incident occurs, then it is beneficial to physically go out to the location at the site and personally follow-up to get a good impression of what is going on to help develop plans to prevent reoccurrence and to complete any required reporting.

Environmental related documents can, and where necessary, should be marked as records so that they are not automatically deleted after 3 years.

Lesson Learned – Site Environmental experts at a laboratory facility should inspect site conditions after an Environmental incident

It can be invaluable for Environmental experts at a laboratory facility, like Shell Technology Center Houston, to get out to the site after Environmental incidents occur. This allows the experts to understand the causes of incidents, conditions under which the incidents occurred, and helps them to determine ways to help the site prevent recurrence of similar incidents. This is not to say that normal Incident Investigations are not warranted, but these visual observations by Environmental staff who understand the regulatory and non-regulatory requirements, gives them additional data for future improvement planning.

2.3. Processes at STCH for compliance and permitting new sources?

The process at STCH for permitting new sources is initiated through the site Management of Change (MOC) process. The MOC HSSE representative involved in individual MOC

reviews makes sure that participants are aware when permitting new sources what would be required by changing site activities. If a new Environmental permit will/could be required, then the MOC process triggers an interface with Ms. Collier. The MOC Coordinator may also bring in other SMEs that can address specific topics in the permit.

MOCs are discussed in weekly meetings, and if those are attended then it is easy to follow any developments on MOCs requiring new Environmental permits or other Environmental activity. The HSSE manager also discusses any ongoing projects during Incident Review Team (IRT+) meetings and in the weekly project status meetings. These internal and external meetings are additional controls to help assure that there is appropriate Environmental team involvement in each actual or possible new permit request or other Environmental need.

2.4. Key Stakeholders at STCH (20+ VP-ships & Multiple Project Groups)

There are representatives in the site HSE team to cover each of the VP-ships at STCH. These HSSE representatives will update Ms. Collier if something new is going on and they will ask her about any future steps that need to be taken from an Environmental perspective.

2.4.1. Project Groups – Capital Projects, Real Estate, 3rd Party Tenants

Third parties are generally brought to STCH through the Real Estate group, as Real Estate facilitates the lease for the third-party tenants. These 3rd Party leases each require an HSE plan, which Ms. Collier helps deliver through an environmental review. For some Environmental items, Environmental services and support are provided on site and for other items the third party will have to secure their own Environmental expertise to help with permit applications, reporting, and general environmental compliance.

Real Estate has several other interfaces that could require engagement with the site Environmental expert. Examples include:

- Real Estate creates a new class 2 waste that needs to be tracked.
- The Landscape team uses herbicides, pesticides and algicides on site and those need to be vetted to make sure they do not create issues for the site. Changes to these would require Environmental engagement.
- Recycling efforts to lead sustainability efforts. For example, ordering furniture can play into sustainability goals for the site and environmental expertise on circular economy issues of new furniture lines may be needed.

Ms. Collier is both a knowledge source for Real Estate, as well as a support partner should they require it. Recently the responsibilities of the Real Estate team have decreased, and

those responsibilities have moved to PTX Operations.

Advice: To keep up to date with changes it is beneficial to maintain close relationships with all parties involved so that there is an awareness if there are any changes. For example, Brown & Root has moved from a Real Estate contract to P&T, so having close connections means that the new interface for that contract is known from the start. Many of these changes have happened in the past year, so there is still an ongoing process around the changes that have been put in place. Managing these organisational changes can only be done through frequent engagements with the parties involved.

Building these relationships helps build and earn trust. If people know what must be done with the information and that it will not be a negative impact on an individual if they share trust. At most Shell locations no one intentionally goes against regulations, so the best thing to do is admit that a mistake has been made. That can then be fixed and make it possible to move forward.

Recommendation for the Future – Staying ahead of Environmental changes at a Shell laboratory have been dependent upon having strong personal relationships with key site stakeholders based upon mutual trust

Roles and responsibilities at a laboratory site like Shell Technology Center Houston (STCH) change with personnel changes, site leadership changes, and site management approaches change with time. The best way to ensure understanding of the possible environmental implications of changes at a site as large and complex as STCH, is to maintain close relationships with all parties involved so that there is an awareness if there are any changes. Building these relationships and not placing blame when things go wrong, helps build and earn trust with key stakeholders. During times of change, these relationships pay dividends by unsolicited sharing of information with the Environmental focal point, and support to ensure seamless handover of activities that help to maintain environmental compliance and improved environmental performance.

2.5. Regulatory Contacts & External Contacts (Key Contacts)

[Limited Time was spent on this topic during the ROCK interview]

TCEQ

There are key contacts in each media. TCEQ has an extensive list on their website. Other than that, TCEQ will contact Ms. Collier as she is listed in their system as the key contact for all associated programmes.

- Camella Widdenhoffer is the contact in TCEQ for the Title 5 permitting process.
- Ms. Toledo works in the subsidence district.

EPA

The EPA process is much the same as TCEQ and they will contact Ms. Collier for inspections etc.. For both EPA and TCEQ, Maurice Pinder will take over the responsibility as the key STCH contact going forward.

[Please review to “Section 8: Personal Contacts” for additional information on external regulatory contacts.]

Shell Internal not STCH

Ms. Collier is connected with SMEs across Shell on waste etc. She is also a member of forums within Shell, such as the Carbon & Environment group and the Waste group. They both have monthly connections that she will attend. However, STCH being a research facility is quite different to most other facilities in Shell (upstream or manufacturing) and STCH therefore has different regulatory requirements compared to most other classic Upstream or Manufacturing locations in the US. The other technology centres in Shell are all in other countries, so they fall under very different regulations, which are not directly comparable to those that apply to STCH.

- Diana Hank, a waste SME, holds a meeting on a regular basis that allows those interested to connect and discuss the different issues around waste and auditing.
- Lynda Beach, Environmental Skillpool manager. She helps bring relevant parties together and helps develop staff.
- Michael Kohnke leads the groundwater group.
- Albert Janssen leads the group for Water and Waste.
- Cindy Chen can be contacted to see who currently manages the latest Air related items. STCH is such a small facility regarding air, Ms. Collier attends the meetings to keep informed of general developments, most of which are not applicable to STCH.
- STCH has a dedicated Shell attorney, Ms. Erin Pena.

3. License to Operate Success

It is often undervalued how much work goes into regulatory compliance.

Challenge for the Future – If Shell Technology Center Houston (STCH) is moved to the Woodcreek site

A challenge for the future would be if the STCH decides to move operations to the nearby Woodcreek site. All current permits would have to be analysed and likely new permits would have to be acquired for such a move. Some of the items of particular concern include: permitting a new flare at Woodcreek (each PBR permit would have to be assessed for the new location as these permit requirements are tied to the distance to the nearest receptor), certain STCH permits (e.g.: the Cogen unit) are grandfathered under older regulations and wouldn't likely remain grandfathered with a location move. Furthermore, environmental compliance would need to continue to be managed at STCH for closure while new permits are being acquired for research related activities at Woodcreek.

Challenge for the Future – Changing activities at Shell Technology Center Houston (STCH) could either simplify or complicate Environmental permits and environmental compliance at the laboratory site

The STCH laboratory site is currently at a major decision crossroad. It is unsure if the STCH will move, if it will stay where it is and get smaller, or if it will stay where it is and take on more 3rd Party customers and grow. If STCH moves, the regulatory requirements will likely increase as historically grandfathered equipment/permits having to be re-permitted for the new location. If STCH gets smaller due to changing activities, some Environmental permitting and compliance elements might become easier. However, if new third parties are brought onsite, they would need to be aligned with STCH's operations at the time, which might require different and/or additional environmental controls.

Risk for the Future – Numerous small Environmental permit violations at a laboratory facility can lead to unwanted attention from regulators. Managing the small things (or the details) is therefore very important

It is often the little things that sometimes get forgotten about, which in turn can cause minor permit violations. Numerous small violations can get the attention of regulatory agencies and bring agency personnel to the site for inspections because they are seeing these numerous small permit violations. This means that if required information or data is omitted from Environmental reports, or if minor permit

infractions occur, these can have detrimental implications down the line. This is because Shell Technology Center Houston is a self-reporting entity, so not including things, or multiple minor violations can cause larger and possibly expensive repercussions for the business if regulators get interested and more involved.

4. Site Specific Topics (Brief History & Current State of Each)

4.1. MB Tank Farm, Contamination

There has been some historical contamination from spills (gasoline, diesel) at the tank farm. There is a report documenting this, which has recently been incorporated into the excavation procedure. There might be new information that needs to be gathered, but that will be determined by future activities planned in the area.

In certain areas the soil is labelled as hazardous waste and in other areas it is not. It depends on the chromium and lead content (not organics) of the soil that was identified by past work and soil testing.

A research group once on site put in some research projects for testing remediation of the soil, which they wanted to apply commercially. For the projects, a site characterisation report was written up, which has been added to the current excavation procedure currently in place.

4.2. Holding Ponds

The dry basin was put in as a storm water retention pond for the installation of one of the new buildings on site (N-building). It was a requirement for construction of that building, so the area needs to be maintained. However, adequate maintenance was not done until recently. With the recent maintenance a concrete pathway was put in to allow water to flow. Care needs to be taken to keep that area well maintained because if not it can quickly become overgrown and inaccessible, thus becoming a wetland.

Advice

It would be beneficial to turn responsibility for maintaining the storm water retention pond area over to Real Estate, a group that already manages similar areas on site. This work is currently done through Compass, a third party that does landscaping services for Real Estate. Many contractors do not like to maintain an area like the dry basin because it remains semi wet (snake habitat, machinery movement issues, it gets overgrown quickly, etc.).

Furthermore, the facility needs to keep removing litter because if not, litter will end up in similar areas and be washed off site (outfalls). It can be difficult to motivate teams to pick up litter that is someone else's garbage, but it must be removed.

New Sandblasting & Painting Area

With the new sandblasting and painting area on site, care still needs to be taken to not execute work if winds are unfavourable for any of the neighbours.

There is no procedure in place yet, but it would be beneficial to do so. If a new painting booth needs to be installed, a PBR will need to be set up specifically for this unit.

Recommendation for the Future – A sandblasting procedure for the new sandblasting area at the Shell Technology Center Houston should be developed and implemented to protect neighbouring properties in certain wind conditions

Highway 6

One of the holding ponds (the first or second pond past the dry basin – pipe is visible on west side) is also connected to a ditch that runs along Highway 6. If there is an accident on the highway, it could potentially lead to liquids from the accident (e.g. petrol or diesel) getting into the holding pond. This connectivity was present even before the site was built up by Shell (previously there was a creek running through the site).

For example, in the past, there was a spill on the highway that potentially caused fish deaths in the pond.

- Gerry Edge – An independent contractor (structural engineer) who manages permitting for STCH with the city of Houston has knowledge about the history of this situation.

4.3. Well #3 Issue, Well #1

Freshwater Well 3

The important point for this well is that the permitted extraction limit is not exceeded (3 million gallons). It is very easy to withdraw that maximum amount.

One of the goals is to keep water in the holding ponds, if those get too dry then the fish in those ponds will die. This could lead to external sources thinking that the cause of the fish death is some chemical that was released in the water, when that is not the cause.

The main use of the well is to put water in the ponds so that the correct water level is maintained. If the extraction limit is reached, then adding more water from the well into the pond is not possible. This rarely happens unless additions are not managed properly.

Freshwater Well 1 & Freshwater Well 2

Wells 1 and 2 were used to supply potable water for the facility. However, the subsidence district halted groundwater withdrawals because of subsidence. Those two wells have been plugged and formally abandoned. However, it is possible to drill those plugs out and the subsidence district has stated in the past that one well can be drilled for emergency use (freeze protection), if desired.

In order to move forward, conversations need to be started up with the subsidence district around what is available for the site and the types of permit volumes would be allowed.

Texas Railroad Commission (TRRC)

To drill an oil or gas well in Texas, one needs to be registered with the Texas Railroad

Commission under the P5 document. Shell had wells in the Permian Basin, outside of College Station, and in Hempstead. Those wells have officially all been plugged and abandoned. However, the TRRC continued sending renewals for the P5, even though those wells were not active anymore. There are two other Shell staff who are familiar with these issues:

- Toylaine Spencer - PCRO (Spencer, Toylaine H SHLUSA-FM/I/P/A1)
- Jane Newitt - (Newitt, Jane A SHLUSA-FM/I/P/A)

5. LOPC Reduction Campaign

STCH does a lot of research for other Shell entities, as well as third parties. Samples are sent via various courier services (FedEx, UPS, DHL, etc.). In the past, many of the samples that came in were leaking likely due to incorrect packaging prior to shipping which may have been completed by folks in the field or even companies who actually prepare samples for shipment. This created a lot of problems for our site, so something needed to be done to improve sample shipping.

New Processes Instituted

If a leaking package came from another Shell entity, then the owner would be looked up in the Shell database and an incident management report would be sent to them, as they were accountable for that shipment (and hence accountable for the spill).

Instituting this accountability has led to a large reduction in sample spills by educating the shippers and attributing spills to them. Those sending improperly shipped packages could not adjust shipping practices until they knew that the packages arrived with leaking liquids.

For third parties, sample packaging guidance has been shared with them. Shell cannot mandate the guidance, but many are now doing a better job at sending samples without leaks.

These process changes have led to a big reduction in leaking packages, and to lower LOPC counts.

Leah Hogendoorn is a Dangerous Good Specialist for the Americas, but she did not want to take on the responsibility for writing the shipping document, as many of the shipping products did not meet dangerous goods requirements. She did provide guidance in publishing the document and has audited STCH in the past.

Lessons Learned – Volumes of Sample Spills were significantly reduced when shipping parties were made aware of improper packaging issues and were held accountable for the sample spill volumes

The Shell Technology Center Houston laboratory does a significant amount of research for various Shell entities, and other 3rd parties. Historically, many of the samples that were shipped to the site via normal courier services (FedEx, UPS, DHL, etc.) arrived leaking. A process was instituted that leaking package entities were contacted about the spill from their samples, were educated on proper shipping procedures, and spill volumes were attributed to their respective organizations, if they were Shell entities. This procedure significantly reduced spill volumes due to improperly shipped samples over time. (Non-Shell entities were sent proper sample shipping guidance when spilled samples were received from them.)

6. Sustainability: Net Zero Emissions, Respect for Nature, etc.

Sustainability goals come from Shell head office, so no selection decisions on these goals are made at STCH. Projects at STCH do not want to spend the money on certain things that might help reach sustainability goals, which means that the Environmental team has to make sure to keep awareness high around sustainability so that goals might be adopted on a project-by-project basis without mandating them. For example, a new project could look to utilise renewable electricity, but the project would have to go out and source it.

An Environmental Specialist cannot influence this process much. The only things that can be done are to stay connected with projects that have sustainability opportunities (e.g., recycling materials during a building demolition), make them aware of Shell Sustainability goals, and require them to report their sustainability data into the Environmental Specialist.

The three key challenges in this space are:

- Implementing sustainable development goals that come from Shell Group are often very difficult because these goals are not aligned with sustainability opportunities that can actually be implemented at the STCH site.
- The Environmental staff at STCH have limited ability to influence projects and business entities on site to incorporate Shell Group sustainability goals into their practices.
- Projects have a limited budget and often do not use that budget to pursue sustainable development goals, or only choose to adopt low-cost sustainability goals.

Challenge for the Future – Shell Group Sustainability Goals are often not applicable to a Shell Research type facility and those that are face significant challenges to achievement at a business or project level at the site

There are 3 key challenges to implementing Shell Group Sustainability goals at a laboratory facility like the Shell Technology Center Houston: Implementing sustainable development goals that come from Shell Group are often very difficult because these goals are not aligned with sustainability opportunities that can actually be implemented at the STCH site. The Environmental staff at STCH have limited ability to influence projects and business entities and on site, and no power to mandate that they incorporate Shell Group sustainability goals into their practices. Projects have a limited budget and often do not use that budget to pursue sustainable development goals, or only choose to adopt low-cost goals.

Respect for Nature

There are a number of activities at STCH that contribute to or could be categorized as Respect for Nature activities:

- The storm water programme is a big contributor to Respect for Nature.
- Assurance activities for recyclers, and disposal companies etc.. Not all recycling and disposal companies are run well and can harm nature in a variety of ways. Waste related assurance activities minimize Shell from using those companies that are poorly run, and whose activities would negatively impact the environment and nature.
- Advising landscaping companies on chemical use on site. This could result in the landscape companies limiting the types and amounts of chemicals used in their activities, guide their activities so that even if certain chemicals are used, they are used in a manner that does not negatively affect other areas (e.g., drift into the water). Note: This requires Environmental staff reviewing the products being used, as some items on the market for commercial use can be very toxic for the environment.
- All the cats on the site have been spayed and neutered so that they do not continue multiplying.
- An attempt was made to incorporate the Wildlife at Work programme at STCH, but due to staff constraints, this program had limited success. This program could be revisited again in the future.

7. Environmental Support Work (Consultation, MOCs, Project Recycling, Respect for Nature, Flaring Issues, Chilled Water Pump)

Environmental support is almost always involved in the MOC process, as each project will typically have at least a small environmental element associated with it. For example, the STCH site transformation project included significant demolition work, which would lead to a lot of opportunity for recycling of waste streams. Environmental support focuses on connecting with the project HSE Manager and support them and SET to make sure that the project would receive the support to be able to recycle as much material as possible.

The flare is a permitted source, a pollution control device, and comes under the highly reactive volatile organic compounds (HRVOC) regulations. For some of the flare projects in addition to just the site Environmental expert, other Shell SMEs have been brought in for their support. Environmental support is always made available for flare related issues via the Environmental Specialist.

There is a biotreatment request form online that all projects have to fill out if the project is going to create wastewater that requires disposal in the process sewer facilities. When a project has completed the form, it is sent to the three different approvers (operational, environmental, and waste). In this way, Environmental support is included in potential changes and/or potential impacts to the POTW or biotreatment plant, downstream. See the waste management procedure found in the HSE Manual for details.

There are so many projects coming and going at STCH, with very diverse waste streams, the site must take care not to upset the publicly owned treatment works (POTW). Because STCH is the only industrial contributor to that local POTW, they will automatically blame STCH if an issue comes up (whether STCH is responsible or not). Environmental support gets involved as requested to help address POTW related issues and prevent problems at the local POTW.

Best Practice – Effective communication and interpersonal skills are critical to Environmental support staff working at a Shell laboratory site such as Shell Technology Center Houston

Communication is one of the best traits someone can have who is working in the environmental space. If there is poor communication then that will put other parties on edge, which can be very detrimental in the environmental space. It is important not to give the impression that you are holding back or hiding something. Open and honest communication, including listening to the other party and trying to understand their view, as well as trying to meet their needs, is of great benefit. Other parties

(regulators etc.) have a reason for doing what they need to do and so letting them do that will put them at ease.

8. Personal Networks

Key Contacts: Internal to Shell

Environmental

- Diana Henk – Senior Soil & Groundwater Specialist
- Michael Kohnke - Soil & Groundwater SME
- Albert Janssen is the Water and Waste SME with PTS/ED
- Cindy Chen is the Waste person in PTS/ED helps with auditing vendors. Shell's representative at CHWMEG. She manages audit requests, which can come at a reduced rate if applied for in advance.
- Tim Smith (Environmental Skillpool) is a Waste SME who analyses the CHWMEG reports. Sometimes those reports are pre-processed by third parties to make them more manageable for the SME to review.
- Environmental Attorney – Erin Pena
- Ellen Hall – Air Emissions Tool – Joe Day can provide #
- Kristin Knoll K2Environmental can score the CHWMEG and 3rd party waste audits prior to SME review if the SME requests this service, her number is 615-579-0727.

STCH Site Contacts

- Site Generator rep. Carlos Gutierrez w/ B&R
- Painting & Sandblasting (SDSs) rep. Jerry McNeil w/ B&R
- Boilers, Cogen, Flare, Gasoline, and Diesel – CJ (Callan Dyke) at Utilities with B&R – 281-544-9304, 832-232-1021
- Refrigerant's rep. James Pianezzola (JP) I think w/ B&R – Refrigerants inventory only accessible on the one computer in HVAC office. 832-474-5241
- Pioneer – Waste (Shredding) Lester Bradley
- Compass – Waste and Recycle (Office and Café) Geramy (Gerald) Roach 832-272-6696 and Daniela Pineda 346-804-9074 and Cristina Gonzalez 713-922-1684
- Stormwater – Dustin Frazier with SET – 713-899-6660
- Potable Water – CJ w/ B&R and Jerry Quintanilla with B&R + Dustin Frazier with SET
- Waste – Luke Pickrell – SET – 281-380-3417
- Brian Joiner – SET for all services assistance – 281-620-3673
- Aziz Meleka – Real Estate Projects Portfolio Manager – 480-329-0536
- Derrick Heyde – American Fire Protection - Fire Back-Flow Preventers
- Brent Norris – Fire Chief – 979-739-6829 Should be listed in SPCC/SWPPP Plans
- Brian Ussery – Site Shell Good Guy that does a lot with projects and helps Environmental a lot – 832-573-3475

- Brian Webster – Site Infrastructure & Utilities Manager 360-333-1556
- Carol Whitmire PTX/T (Knows a lot about that group, been here 30+ years) – 281-544-8726, 832-202-6223
- Charlie Solito – RE Environmental guy – 281-544-9291, 907-280-7812 – Will help you gather and can provide you the RE Project waste
- Dan Didier – SET's Regulatory guy if you need any help, mostly in waste 713-641-7574
- Jaime Smith – Electrician for Transformer questions or the working of the Stationary Generators all over site – 713-412-2519
- Jerry McNeil – B&R Safety and posts Paint & Sandblast records 281-636-8249
- Jerry or Gerry Quintanilla – knows a lot about the STCH plumbing system and will come to you with repair on the potable water system work – you can ask him about disinfection 281-492-4119
- Joe McKinney – SET – good waste contact and was at STCH for a while 713-816-3728
- STCH Maintenance 713-241-2345
- Meagan Middleton – SET – 281-253-2513
- Mike Fish Micro-Grid – Nat Gas Engine and renewable power generated
- Randy Neuendorff – B&R 979-942-6838 – he can help with hanging stuff at gate for permit public notice and inspects some stormwater stuff around their activities
- Rekha Hanumanthan – SET provides numbers for Quarterly Environmental report
- Robert Hardy – B&R works on potable water and in soils/excavations, has been around a long time – 832-310-3911
- Trung Vo – Pi guy for flare info. 409-225-1889
- Rosa Aguilar – EO Micro Reactors – 281-544-6931
- Unit 16 (After Hrs Guys that rove around site. They do initial response to releases at night that they find so they may call you. Good to give them your Shell number 832-416-2037
- Vehicle Assistance - @ STCH if your car won't start, etc. 281-544-8155
- Victor Maldonado – B&R and has been here a long time. Knows a lot about a lot of things but has his finger on the pulse of back-flow preventers 713-498-5606

Other

- Computer Help Desk – Shell – 312-843-5291

Key Contacts: External

- Yolanda Toledo – 281-956-2189 Hours: 6 – 3 Harris-Galveston Subsidence District (HGSD)
- Camilla Widenhofer – TCEQ Federal Title V Permit Engineer

- TCEQ Air Permits Division
P.O. Box 13087, MC 163
Austin, TX 78753
Phone: (512)239-1028
Fax: (512)239-1300
camilla.widenhofer@tceq.texas.gov
- TCEQ Reg 12 Air Guy – 713-767-3720
- TCEQ Air others 713-767-3712, 713-767-3500, 713-767-3714
- TCEQ – All water issues WQ #0001853000 888-777-3186
- TCEQ – Austin 512-239-1000 or 512-463-7727
- TCEQ – Spill # 713-767-3563
- TCEQ STEERS 512-239-6925
- Harris County Pollution Control – 713-920-2831
- NRC – Coast Guard Command Center to report RQ/reportable releases other than minor stormwater releases – 866-370-3865
- City of Houston Air Monitoring Trailer by the MicroGrid – 832-393-5703, Contact James Fabian 281-865-6813 - Combination for lock either for them or our landscape guys to get in there to mow is 9898
- City of Houston Potable water lady, Julie Huerta 346-266-4483
- City of Houston guy that flushes the hydrant coming inside STCH, like for low chlorine residual – Scott Barker 713-562-2957
- City Potable Water Lab for Bacterial Testing – 832-393-3939 – Our account is #1010557
- Terri May – AECOM (Air Consultant) Cell – 832-875-7351 best way to reach her Katy Office 281-675-3591
- Angela Johnson – AECOM (Waste/SPCC/SWPPP Consultant) – 832-244-2903
- XENCO Lab now Eurofins Debbie Simmons 281-240-4200 Ext. 1104 and cell 281-630-9242 if an emergency – Does analysis on water and waste for us.
- Eurofins – Lance Tigrett if you need help with issues on analysis 979-484-9088

9. Best Practices

Best Practices	Additional Metadata (in Addition to Document Metadata)
Best Practice – Environmental Training is more effective when done in person for on-site staff that could impact storm water discharge limit compliance When training staff and contractors at a laboratory facility like Shell Technology Center Houston, it is important to help them understand that they are an important part of the process in keeping the storm water conveyance system as clean as possible. Each activity on site can cause issues for the storm water system if precautions are not taken. It is therefore important to follow the Storm Water Pollution Prevention Plan and make sure there is good communication (best achieved through in-person visits) with the various parties working on site.	Social Performance, Community Benefits, Social Impacts
Best Practice – Environmental compliance requirement monitoring is improved through use of a central tracking tool All laboratory site Environmental compliance requirements (i.e.: testing, reporting, training) at Shell Technology Center Houston are managed together in one single electronic repository. (i.e.: An Excel spreadsheet) Centralizing this information for the requirements of all environmental media (Air, Water, Waste) helps ensure that compliance oversight is consolidated and can be monitored/managed in one place. Distributing these requirements to various departments or individuals leads to higher exposure for non-compliance during staff/organizational changes.	

Best Practice – Effective communication and interpersonal skills are critical to Environmental support staff working at a Shell laboratory site such as Shell Technology Center Houston Communication is one of the best traits someone can have who is working in the environmental space. If there is poor communication then that will put other parties on edge, which can be very detrimental in the environmental space. It is important not to give the impression that you are holding back or hiding something. Open and honest communication, including listening to the other party and trying to understand their view, as well as trying to meet their needs, is of great benefit. Other parties (regulators etc.) have a reason for doing what they need to do and so letting them do that will put them at ease.	

10. Lessons Learned

Lessons Learned	Additional Metadata (in Addition to Document Metadata)
Lesson Learned – Site Environmental experts at a laboratory facility should inspect site conditions after an Environmental incident It can be invaluable for Environmental experts at a laboratory facility, like Shell Technology Center Houston, to get out to the site after Environmental incidents occur. This allows the experts to understand the causes of incidents, conditions under which the incidents occurred, and helps them to determine ways to help the site prevent recurrence of similar incidents. This is not to say that normal Incident Investigations are not warranted, but these visual observations by Environmental staff who understand the regulatory and non-regulatory requirements, gives them additional data for future improvement planning.	
Lessons Learned – Volumes of Sample Spills were significantly reduced when shipping parties were made aware of improper packaging issues and were held accountable for the sample spill volumes The Shell Technology Center Houston laboratory does a significant amount of research for various Shell entities, and other 3rd parties. Historically, many of the samples that were shipped to the site via normal courier services (FedEx, UPS, DHL, etc.) arrived leaking. A process was instituted that leaking package entities were contacted about the spill from their samples, were educated on proper shipping procedures, and spill volumes were attributed to their respective organizations, if they were Shell entities. This procedure significantly reduced spill volumes due to improperly shipped	

samples over time. (Non-Shell entities were sent proper sample shipping guidance when spilled samples were received from them.)	
---	--

11. Risks / Challenges for the Future

Risks / Challenges and Recommendations for the Future	Additional Metadata (in Addition to Document Metadata)
---	--

Challenge for the Future – If Shell Technology Center Houston (STCH) is moved to the Woodcreek site A challenge for the future would be if the STCH decides to move operations to the nearby Woodcreek site. All current permits would have to be analysed and likely new permits would have to be acquired for such a move. Some of the items of particular concern include: permitting a new flare at Woodcreek (each PBR permit would have to be assessed for the new location as these permit requirements are tied to the distance to the nearest receptor), certain STCH permits (e.g.: the Cogen unit) are grandfathered under older regulations and wouldn't likely remain grandfathered with a location move. Furthermore, environmental compliance would need to continue to be managed at STCH for closure while new permits are being acquired for research related activities at Woodcreek.	
Challenge for the Future – Changing activities at Shell Technology Center Houston (STCH) could either simplify or complicate Environmental permits and environmental compliance at the laboratory site The STCH laboratory site is currently at a major decision crossroad. It is unsure if the STCH will move, if it will stay where it is and get smaller, or if it will stay where it is and take on more 3rd Party customers and grow. If STCH moves, the regulatory requirements will likely increase as historically grandfathered equipment/permits having to be re-permitted for the new location. If STCH gets smaller due to changing activities, some Environmental permitting and compliance elements might become easier. However, if new third parties are brought onsite, they would need to be aligned with STCH's operations at the time, which might require different and/or additional environmental controls.	

Risk for the Future – Numerous small Environmental permit violations at a laboratory facility can lead to unwanted attention from regulators. Managing the small things (or the details) is therefore very important It is often the little things that sometimes get forgotten about, which in turn can cause minor permit violations. Numerous small violations can get the attention of regulatory agencies and bring agency personnel to the site for inspections because they are seeing these numerous small permit violations. This means that if required information or data is omitted from Environmental reports, or if minor permit infractions occur, these can have detrimental implications down the line. This is because Shell Technology Center Houston is a self-reporting entity, so not including things, or multiple minor violations can cause larger and possibly expensive repercussions for the business if regulators get interested and more involved.	
Challenge for the Future – Shell Group Sustainability Goals are often not applicable to a Shell Research type facility and those that are face significant challenges to achievement at a business or project level at the site There are 3 key challenges to implementing Shell Group Sustainability goals at a laboratory facility like the Shell Technology Center Houston: Implementing sustainable development goals that come from Shell Group are often very difficult because these goals are not aligned with sustainability opportunities that can actually be implemented at the STCH site. The Environmental staff at STCH have limited ability to influence projects and business entities and on site, and no power to mandate that they incorporate Shell Group sustainability goals into their practices. Projects have a limited budget and often do not use that budget to pursue sustainable development goals, or only choose to adopt low-cost goals.	

12. Opportunities for Improvement / Recommendations for the Future

Opportunities for Improvement / Recommendations for the Future	Additional Metadata (in Addition to Document Metadata)
Recommendation for the Future – Staying ahead of Environmental changes at a Shell laboratory have been dependent upon having strong personal relationships with key site stakeholders based upon mutual trust Roles and responsibilities at a laboratory site like Shell Technology Center Houston (STCH) change with personnel changes, site leadership changes, and site management approaches change with time. The best way to ensure understanding of the possible environmental implications of changes at a site as large and complex as STCH, is to maintain close relationships with all parties involved so that there is an awareness if there are any changes. Building these relationships and not placing blame when things go wrong, helps build and earn trust with key stakeholders. During times of change, these relationships pay dividends by unsolicited sharing of information with the Environmental focal point, and support to ensure seamless handover of activities that help to maintain environmental compliance and improved environmental performance.	
Recommendation for the Future – A sandblasting procedure for the new sandblasting area at the Shell Technology Center Houston should be developed and implemented to protect neighbouring properties in certain wind conditions	

13. Key Information Sources

None noted during the ROCK interview, but several were shared in handover work with those that will be taking on Ms. Collier's work responsibilities. (Maurice Pinder and others)

Literature References

None noted during the ROCK interview.

Bibliographic Information

Report Number	SRN-010382
Title	Retention of Critical Knowledge - Beth Collier – Environmental Specialist: Air, Water and Waste - GSUSI-PTS-S
Subject	Retention of Critical Knowledge Report
Keywords	Environmental Compliance; Site Environmental Support; Shell Technology Center Houston; STCH; Laboratory; Environmental Compliance; Air Compliance; Water Compliance; Waste Compliance; Air Emission Management; Water Discharge Management; Waste Management; Air; Water; Waste; Respect for Nature; Sustainability; Critical Knowledge; Knowledge Retention; LFE; Learning from Experience
Author(s)	Kosiara, Mark SHLUSA-HRL/LK; Collier, Beth B GSUSI-PTS/S
Reviewed by	Smith, Rob L GSUSI-PTS/ST
Approved by	Smith, Rob L GSUSI-PTS/ST
Issuing Company	Shell USA, Inc.
Publishing Date	September 2024
Security Classification	Internal